

JUCO: A Joint Uncertainty-Calibrated Optimisation Framework for ICU Mortality Prediction with Safe Abstention Under Informative Missingness

Avoy Nath Chowdhury, Anisha Kumari, Prasun Tripathi

Abstract—Accurate prediction of in-hospital mortality for ICU patients enables timely clinical intervention, yet deployment across institutions is undermined by the outcome-dependent missingness endemic to electronic health records (EHR). Because clinicians order laboratory tests in proportion to their suspicion of patient deterioration, the absence of a measurement is itself a Missing-Not-At-Random (MNAR) signal that conventional imputation discards. To the best of our knowledge, no existing framework simultaneously quantifies imputation uncertainty, propagates it structurally to a downstream classifier, and converts it into principled abstention. We introduce JUCO (Joint Uncertainty-Calibrated Optimisation), a four-phase framework that treats imputation uncertainty as a first-class quantity throughout the pipeline. It consists of LightGBM quantile regression whose prediction interval width encodes epistemic confidence, a proportional trapezoidal fuzzy transformation, a 180-tree soft-routing decision forest in which uncertain samples distribute their weight across branches in inverse proportion to confidence, and a Shannon-entropy abstention protocol that defers high-uncertainty cases to clinical review. All five governing parameters are jointly optimised via two-stage Differential Evolution on MIMIC-IV (82,785 stays) dataset and frozen for external validation on eICU (162,136 stays, 208 hospitals) dataset. Under escalating MNAR stress from 10% to 70%, JUCO’s AUROC (Area Under the Receiver Operating Characteristic Curve) degrades by only -0.023 versus -0.054 for XGBoost and -0.056 for MissForest+XGB, overtaking both at the most severe rate; abstention reduces AURC (Area Under the Risk-Coverage Curve) 2.8-fold (0.032 versus 0.090); and the AUROC gap with gradient boosting narrows by 46% under cross-institutional shift (-0.028 on MIMIC-IV to -0.015 on eICU). These results suggest that carrying imputation uncertainty through each stage of the pipeline, rather than discarding it at the imputation step, improves robustness in multi-site deployment. Code for the JUCO framework, including all four phases, Differential Evolution optimisation, and evaluation scripts, is available at <https://github.com/Avoy01/JUCO.git>.

Index Terms—Electronic health records, external validation, ICU mortality prediction, MIMIC-IV, missing not at random, selective prediction, uncertainty quantification.

A. N. Chowdhury is with the School of Computer Engineering, KIIT Deemed to be University, Bhubaneswar, Odisha, India (e-mail: avoy-nath2004@gmail.com).

A. Kumari is with the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, Uttar Pradesh, India (e-mail: anisha.mishra@mnnit.ac.in).

P. Tripathi is with the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, Uttar Pradesh, India (e-mail: p.c.tripathi@mnnit.ac.in).

Corresponding author: Anisha Kumari (e-mail: anisha.mishra@mnnit.ac.in).

I. INTRODUCTION

Predicting in-hospital mortality for ICU patients is a crucial task in clinical informatics. An accurate estimate at admission can redirect monitoring resources, expedite goals-of-care discussions, and trigger interventions before the patient’s condition deteriorates irreversibly. Gradient boosting models trained on Electronic Health Records (EHR) have achieved AUROC values of $0.82 - 0.88$ on retrospective MIMIC-III/IV cohorts, performance comparable to established severity scores like APACHE and SOFA [1], [2]. Meanwhile, a meta-analysis of 40 studies has confirmed gradient boosting as a leading method (pooled AUROC of 0.83) while identifying model calibration and robustness during deployment as key unresolved limitations [3]. Translation from retrospective benchmarks to prospective multi-site deployment remains infeasible because of a structural data-quality problem: pervasive, outcome-correlated missingness in EHR data.

In the ICU, the absence of a measurement is not an administrative error but a clinical signal: a clinician orders a test in proportion to suspected deterioration, so the absence of a lactate value, for instance, often indicates that hypoperfusion was not suspected. Rubin [4] formalised this as Missing-Not-At-Random (MNAR); under MNAR, the missingness pattern itself functions as a latent predictor of mortality, one whose direction shifts when hospital protocols, staffing, or case-mix differ between training and deployment sites. Models that encode this pattern as a fixed signal become brittle across institutions, a fragility confirmed by evidence that internally validated clinical risk models degrade systematically in external cohorts [5]–[8]. Existing strategies can be divided into two families with the same underlying gap. Point-imputation methods (mean imputation, MICE, MissForest) discard all estimation uncertainty [9], [10], while native NaN routing in gradient boosting [11], [12] implicitly encodes MNAR structure as a proxy variable that is brittle whenever the missingness mechanism at deployment deviates from training [13]. Neither family quantifies imputation uncertainty, propagates it structurally into the classifier, or defers predictions whose quality is compromised by missing key inputs. This gap represents a safety failure mode rather than merely a calibration shortfall in a setting where a confident estimate built on entirely imputed laboratory values gives clinicians false certainty.

We introduce **JUCO** (Joint Uncertainty-Calibrated Optimisation), a four-phase framework (as shown in Fig. 1) that carries imputation uncertainty forward through the

prediction pipeline through quantile-regression intervals, a trapezoidal fuzzy encoding, a soft-routing decision forest, and entropy-based abstention, instead of discarding it before classification. All five governing parameters are jointly optimised via two-stage Differential Evolution on MIMIC-IV dataset.

Main contributions of this study are as follows.

- 1) We propose JUCO, a four-phase pipeline that structurally propagates quantile-regression imputation uncertainty through a trapezoidal fuzzy transformation and a soft-routing decision forest to an entropy-driven clinical abstention protocol, with all governing parameters jointly optimised via two-stage Differential Evolution in a unified end-to-end objective.
- 2) We demonstrate that JUCO maintains substantially greater robustness under escalating MNAR missingness than both XGBoost and MissForest+XGB, while overtaking both baselines under severe missingness conditions.
- 3) Through ablation analysis, we show that the safe abstention protocol substantially improves selective reliability, while fuzzy soft routing independently improves calibration relative to crisp thresholding.
- 4) We provide a mechanistic explanation for JUCO’s lower nominal AUROC under standard conditions, showing that gradient boosting partially derives its advantage from exploiting stationary missingness proxies rather than modelling robustness, an effect that contracts under external validation.
- 5) We validate JUCO on the multi-institutional eICU cohort using parameters frozen from a single training site, demonstrating stable calibration and selective reliability without recalibration across hospitals.

The remainder of this paper is organised as follows: Section 2 surveys related work; Section 3 formalises the problem; Section 4 describes the JUCO architecture; Section 5 presents the experimental setup; Section 6 reports results; Section 7 discusses findings and limitations; and Section 8 concludes. This study adheres to the TRIPOD+AI reporting guidelines for clinical prediction models [14].

II. RELATED WORK

Missingness in ICU EHR data is clinically determined rather than administrative: the decision to order a lab is correlated with its value and with the outcome, producing the MNAR mechanism formalised by Rubin [4] and violating the ignorability assumption on which most imputation methods rely [6]. Gradient boosting (XGBoost, LightGBM) is the dominant modality for ICU mortality prediction, reaching pooled AUROC 0.83–0.92 via native NaN routing that treats absence as a learnable feature [1], [11], [15], with 24-hour-updated dynamic models reaching 0.909 under successful external validation [16]; this is effective within a stable institutional pipeline but degrades when charting protocols, equipment, or case-mix shift between training and deployment [7], [17], a fragility this paper quantifies directly (XGBoost Δ AUROC = -0.054 at 70% MNAR versus JUCO’s -0.023) and that

most published external-validation studies are underpowered to detect [18].

Point-estimate imputation (mean, MICE [9], MissForest [10]) discards estimation uncertainty entirely and offers no universally optimal choice across missingness mechanisms [19]; deep generative alternatives (GAIN [20], MNAR-VAEs [21], recurrent and diffusion imputers [22], [23]) offer no consistent advantage under MNAR and are prone to mode collapse [24]. Uncertainty-aware recurrent imputation networks for clinical time series [25] and granular fuzzy models of incomplete data [26] each address one half of this problem, quantifying imputation uncertainty or representing incompleteness with fuzzy sets respectively, but neither propagates that uncertainty into a downstream classifier’s routing or abstention decisions, the gap JUCO addresses directly. Quantile regression [27] instead yields an interval whose width directly encodes uncertainty without distributional assumptions; to our knowledge this is the first work to propagate such intervals through a soft-routing fuzzy classifier. Trapezoidal fuzzy numbers [28], [29] and soft-routing trees [30] have established theoretical grounding, and the epistemic/aleatoric distinction of Hüllermeier and Waegeman [31] motivates using interval width as a soft-routing signal, but prior work, most recently fuzzy probability trees for misdiagnosis reduction [32], has not jointly optimised imputation and routing at EHR scale.

Selective prediction with a reject option [33], [34], evaluated via AUROC [35] within the deep learning framework [36], uses entropy proven asymptotically optimal under a well-calibrated posterior [37] and consistently improves clinical system-level accuracy [38], [39], including under distributional shift via conformal methods [40]. Prior abstention criteria, however, are based on classification-margin uncertainty rather than input-feature reliability; JUCO instead links abstention mechanistically to imputation quality via the chain wide interval $\rightarrow$ wide trapezoid $\rightarrow$ reduced split sharpness (Eq. 11) $\rightarrow$ diffuse leaf distribution $\rightarrow$ high output entropy. Finally, cross-institutional transportability of clinical prediction models is a well-documented, only partially understood failure mode [5]: supervised learning from incomplete data is consistent only when the missingness mechanism matches at train and test time [7], informative-missingness proxies entangle models with source-institution ordering habits [17], and this “missingness shift” [8] has a crossover point within the range of shifts seen between real institutions [15], [41].

To the best of our knowledge, no prior framework has tested whether uncertainty-propagating representations transfer more stably than missingness-direction proxies; the 46% contraction in JUCO’s AUROC gap from MIMIC-IV to eICU ($-0.028 \rightarrow -0.015$) is the primary empirical test of this claim. Collectively, existing frameworks either eliminate imputation uncertainty (point imputation) or exploit it as a static proxy (NaN routing), and existing abstention frameworks act on classification margin rather than input reliability; JUCO unifies both within a single jointly optimised pipeline.

III. PROBLEM FORMULATION

Each observation corresponds to a single ICU admission, characterised by a feature vector aggregated over the first 24

TABLE I: Summary of notation.

Symbol	Definition
$\mathbf{X} \in (\mathbb{R} \cup \{\text{NaN}\})^{n \times d}$	Feature matrix with informative missingness
$\mathbf{M} \in \{0, 1\}^{n \times d}$	Missingness mask ($M_{ij} = 1$ iff missing)
$\mathbf{y} \in \{0, 1\}^n$	Binary mortality labels
$\alpha_L \in (0, 0.5)$	Lower quantile level
$\alpha_U \in (0.5, 1)$	Upper quantile level
$q_{ij}^L, q_{ij}^M, q_{ij}^U$	Imputed lower, median, upper bounds
$\eta \in [0, 0.5]$	Fuzzy proportion factor
$(a_{ij}, b_{ij}, c_{ij}, d_{ij})$	Trapezoidal fuzzy number for (i, j)
$\boldsymbol{\theta} = (\alpha_L, \alpha_U, \eta, d_{\max}, n_{\min})$	DE-optimised parameter vector
$d_{\max}$	Maximum tree depth in the fuzzy forest
$n_{\min}$	Minimum samples required at a leaf node
$w_k, k \in \{0, 1\}$	Inverse-frequency class weight for outcome class k
H_i	Shannon entropy of predicted probabilities
τ	Entropy abstention threshold
PICP	Prediction Interval Coverage Probability
MPIW _{norm}	Normalised Mean Prediction Interval Width
AURC	Area Under the Risk-Coverage Curve
Risk@90	Error rate on the most confident 90%

hours, a binary missingness indicator, and a binary mortality outcome: $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{M}_i, y_i)\}_{i=1}^n$, where n is the number of ICU admissions in the cohort, $\mathbf{x}_i \in (\mathbb{R} \cup \{\text{NaN}\})^d$ with d the number of features per admission, $\mathbf{M}_i \in \{0, 1\}^d$ with $M_{ij} = 1$ iff feature j is missing for patient i , and $y_i \in \{0, 1\}$ denotes the observed in-hospital mortality outcome, with $y_i = 1$ if the patient died and $y_i = 0$ otherwise. Unlike most clinical prediction frameworks, which treat $\mathbf{M}_i$ as a nuisance resolved by imputation, JUCO retains it throughout all four phases to modulate how much influence each imputed entry exerts on the downstream prediction. Table I provides a summary of all the notation used in this paper.

According to Rubin [4], the ICU missingness mechanism is characterised by $P(\mathbf{M} \mid \mathbf{X}_{\text{full}}, \mathbf{y})$ (Eq. 1), where $\mathbf{X}_{\text{full}} \in \mathbb{R}^{n \times d}$ denotes the complete, fully-observed feature matrix that would have been recorded in the absence of any missingness, with x_{ij}^{true} its (i, j) entry. This mechanism is not ignorable: the order of lactate measurement reflects the suspicion of hypoperfusion, so

$$P(\mathbf{M} \mid \mathbf{X}_{\text{full}}, \mathbf{y}) \quad (1)$$

for entry (i, j) ,

$$P(M_{ij} = 1 \mid x_{ij}^{\text{true}}, \mathbf{x}_{i,-j}, y_i) \neq P(M_{ij} = 1 \mid \mathbf{x}_{i,-j}, y_i), \quad (2)$$

where $\mathbf{x}_{i,-j}$ denotes patient i 's feature vector with feature j excluded. I.e., if the true value of x_{ij} is known, the probability of it being missing changes even after conditioning on all other observed characteristics and outcomes. This is what makes ignorability-based imputation inconsistent and turns the pattern of missingness itself into a latent predictive signal that becomes fragile when the laboratory ordering protocol, equipment, or case-mix changes during implementation.

Given this framework, the goal is not only to accurately predict mortality in the training institution but also to have that accuracy degrade gracefully as the missingness mechanism changes, and to clearly identify erroneous predictions due to high imputation uncertainty.

Problem Statement (Uncertainty-Aware Mortality Prediction Under MNAR). *Given training data $\mathcal{D}_{\text{train}}$ drawn from a distribution satisfying Eq. 2, learn a selective prediction function*

$$f : (\mathbb{R} \cup \{\text{NaN}\})^d \rightarrow [0, 1] \cup \{\perp\}, \quad (3)$$

where $\hat{p}_i \in [0, 1]$ is the predicted mortality risk for patient i and $\perp$ indicates abstention, such that (i) predicted probabilities are well calibrated on retained cases; (ii) abstention concentrates on high imputation-uncertainty cases; and (iii) $\text{AUROC}(f, r)$ degrades slowly as the additional MNAR rate r increases.

These three conditions are stated in deliberate priority order: calibration on retained cases first, because an unreliable probability where the model expresses confidence is more harmful than lower discrimination; abstention quality (AURC) second, because it quantifies what a clinician actually experiences on the predictions they receive; and slow AUROC degradation under missingness shift third, because a model at 0.84 in one institution and 0.78 in another provides less value than one at 0.81 in both.

IV. METHODOLOGY

Figure 1 illustrates the four-phase pipeline. The five governing parameters $\boldsymbol{\theta} = (\alpha_L, \alpha_U, \eta, d_{\max}, n_{\min})$ are optimised jointly via two-stage Differential Evolution (Section IV-C1) rather than tuned independently, so imputation width and routing sensitivity are co-adapted to the same risk-coverage objective.

A. Phase 1: Adaptive Quantile Imputation

For each feature j with missing entries, a LightGBM quantile regressor predicts the conditional quantiles of x_{ij} given the remaining observed features $\mathbf{x}_{i,-j}$, minimising the weighted pinball loss

$$\hat{q}_j^\alpha(\mathbf{x}_{-j}) = \arg \min_f \sum_{i: M_{ij}=0} w_i \rho_\alpha(x_{ij} - f(\mathbf{x}_{i,-j})), \quad (4)$$

where $\rho_\alpha(u) = u(\alpha - \mathbf{1}_{u < 0})$ is the pinball loss [42] and w_i are the inverse-frequency class weights defined in Appendix A, which give deceased patients roughly ten times the weight of survivors at MIMIC-IV's $\sim 10\%$ mortality prevalence. Three regressors per feature, at quantile levels $\alpha \in \{\alpha_L, 0.5, \alpha_U\}$, yield the triple $(q_{ij}^L, q_{ij}^M, q_{ij}^U)$; α_L and α_U are not fixed but form part of $\boldsymbol{\theta}$, optimised jointly by DE (Section IV-C1) rather than committed to a fixed nominal coverage such as 90% or 95%. Observed entries pass through unchanged.

B. Phase 2: Proportional Fuzzy Transformation

A wide interval (poorly constrained value) should soften tree routing; a narrow interval should harden it toward crisp behaviour. A trapezoidal fuzzy number [43] satisfies this because its support width $d - a$ is a continuous, monotone function of interval width, a natural carrier for the epistemic uncertainty of Hüllermeier and Waegeman [31]. Each interval is mapped to $\tilde{x}_{ij} = (a_{ij}, b_{ij}, c_{ij}, d_{ij})$ with membership function

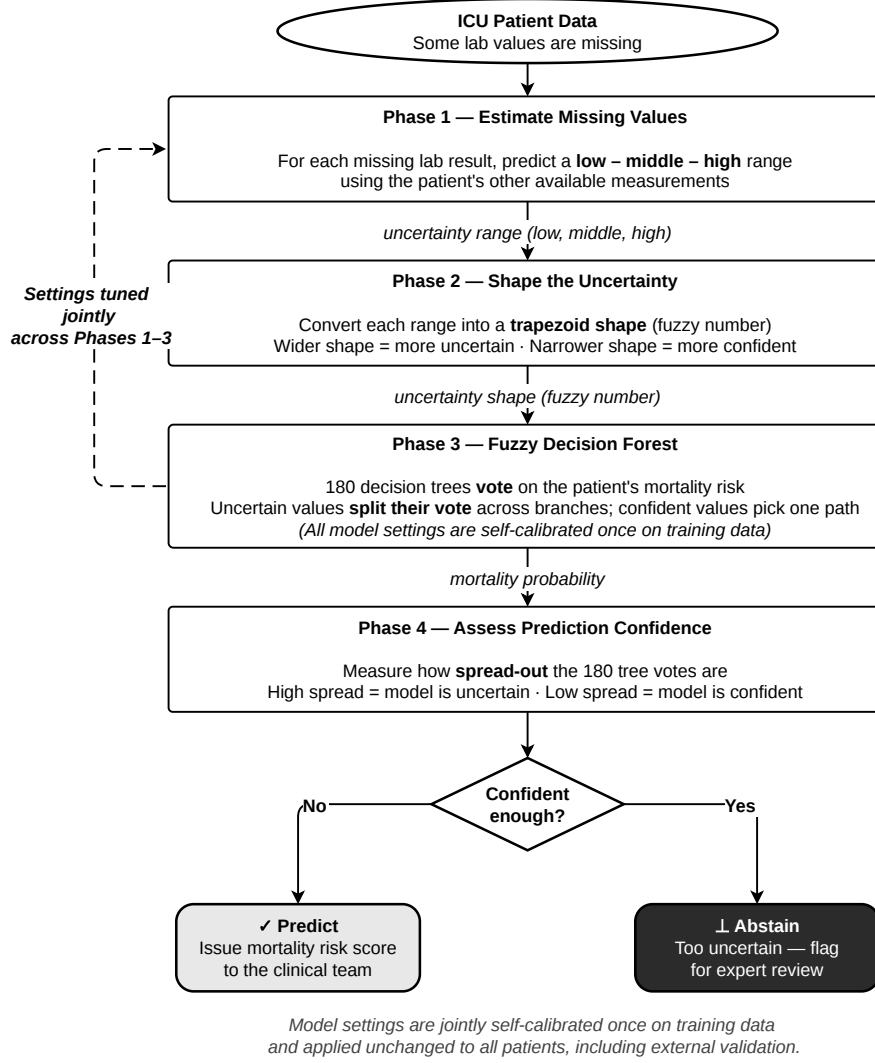


Fig. 1: Overview of the JUCO four-phase pipeline. Phase 1 estimates each missing lab value as a low–middle–high uncertainty range. Phase 2 converts that range into a trapezoid shape whose width encodes uncertainty. Phase 3 routes the patient through 180 decision trees, with uncertain values splitting their vote across branches to produce a mortality probability; settings across Phases 1–3 are jointly calibrated once on training data and applied unchanged thereafter. Phase 4 assesses the spread of those votes: if the model is confident enough it issues a risk score, otherwise it abstains and flags the case for clinical review.

$$\mu_{\tilde{x}_{ij}}(x) = \begin{cases} 0, & x < a_{ij} \text{ or } x > d_{ij}, \\ \frac{x - a_{ij}}{b_{ij} - a_{ij}}, & a_{ij} \leq x < b_{ij}, \\ 1, & b_{ij} \leq x \leq c_{ij}, \\ \frac{d_{ij} - x}{d_{ij} - c_{ij}}, & c_{ij} < x \leq d_{ij}. \end{cases} \quad (5)$$

constructed from the quantile outputs as

$$a_{ij} = q_{ij}^L, \quad d_{ij} = q_{ij}^U, \quad (6)$$

$$b_{ij} = \max(a_{ij}, q_{ij}^M - \eta \cdot (q_{ij}^U - q_{ij}^L)), \quad (7)$$

$$c_{ij} = \min(d_{ij}, q_{ij}^M + \eta \cdot (q_{ij}^U - q_{ij}^L)). \quad (8)$$

where $\eta \in [0, 0.5]$ (part of θ) controls the ratio of the flat core $[b, c]$ to the total support $[a, d]$: small η gives a wide

plateau appropriate under high uncertainty; large η approaches a triangular, near-point-imputation shape. Observed entries degenerate to a crisp point ($a = b = c = d = x_{ij}$). Two scalars carry forward to Phase 3 and the DE fitness function:

$$\text{centroid}(\tilde{x}_{ij}) = \frac{a_{ij} + b_{ij} + c_{ij} + d_{ij}}{4}, \quad (9)$$

$$\text{uncertainty}(\tilde{x}_{ij}) = d_{ij} - a_{ij}. \quad (10)$$

The centroid is the effective feature value for tree routing; it is this uncertainty that drives the sharpness function below.

C. Phase 3: Fuzzy Decision Forest

Each internal node splits the samples using soft membership degrees instead of a strict binary split. For a node splitting on feature j with triangular fuzzy threshold $\mathcal{T} =$

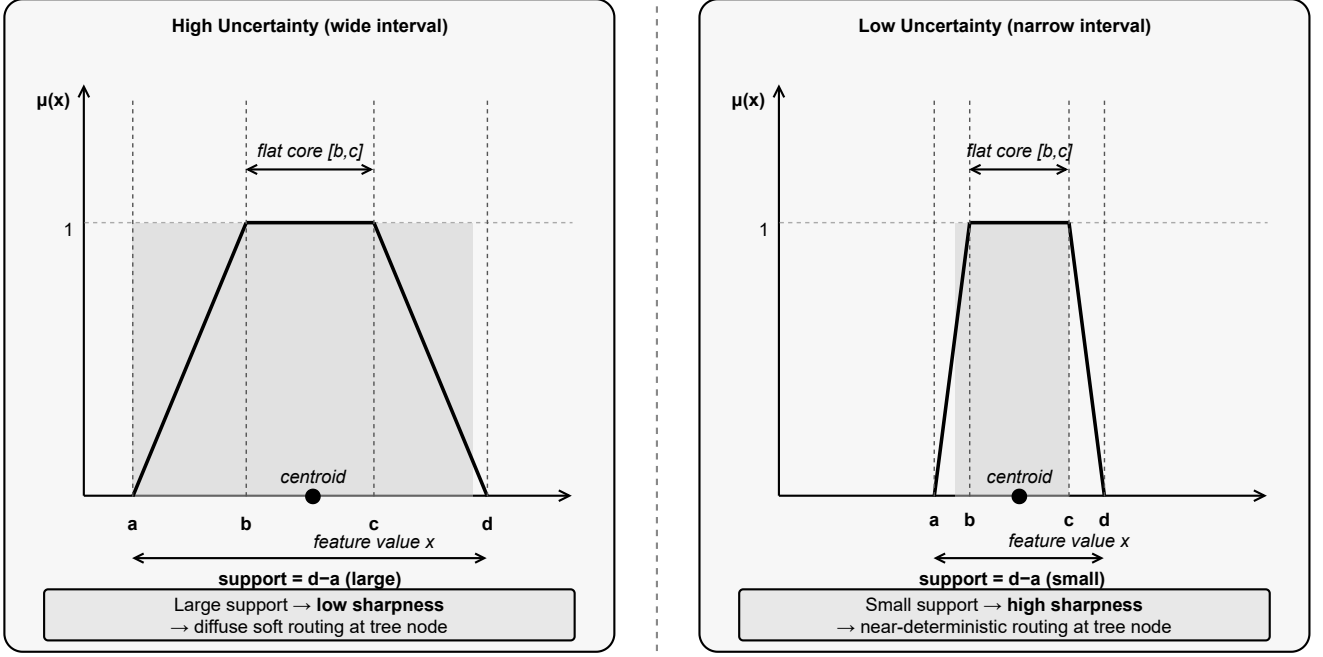


Fig. 2: Proportional fuzzy transformation (Phase 2). A missing entry with a wide quantile interval (high imputation uncertainty, left panel) produces a broad trapezoidal fuzzy number with large support width $d - a$, resulting in low sharpness and diffuse soft routing at the tree node. A narrow interval (low uncertainty, right panel) produces a nearly-clear trapezoid with small support, which provides high sharpness and nearly-deterministic routing. The centroid serves as an effective feature value for tree routing; the support width is the uncertainty signal passed to the sharpness function. The proportional factor η controls the ratio of the flat core $[b, c]$ to the total support $[a, d]$; the observed entries converge to a point: $a = b = c = d$.

$(t_{\text{center}}, t_{\text{width}})$, the left-child membership μ_{ij}^L is a sigmoid, $\sigma(z) = 1/(1 + e^{-z})$, whose sharpness adapts inversely to combined uncertainty.

$$\text{sharpness}_{ij} = \frac{5}{\text{uncertainty}(\tilde{x}_{ij}) + t_{\text{width}} + \varepsilon}, \quad (11)$$

$$\mu_{ij}^L = \sigma(\text{sharpness}_{ij} \cdot (t_{\text{center}} - \text{centroid}(\tilde{x}_{ij}))), \quad (12)$$

with $\mu_{ij}^R = 1 - \mu_{ij}^L$ and $\varepsilon = 10^{-12}$. Observed inputs ($d_{ij} = a_{ij}$, $t_{\text{width}} \approx 0$) drive sharpness toward infinity, producing near-deterministic splits indistinguishable from a crisp tree; imputed inputs with wide intervals produce low sharpness and split their weight across both children, diminishing their leaf influence in proportion to imputation uncertainty. Splits are chosen to maximise fuzzy weighted information gain (fuzzy entropy and gain criteria given in Appendix A), evaluated at seven percentiles of the centroid distribution per candidate feature. At each leaf, the class distribution is formed from the accumulated soft-routed weight vector, so a leaf visited predominantly by imputed samples is more diffuse (and higher-entropy) than one visited predominantly by observed samples, providing the signal Phase 4 converts into abstention.

The forest comprises $K = 180$ trees, each trained on a class-balanced bootstrap sample with a fixed random 70% feature subset per tree; the final prediction averages class probabilities

across all trees and is post-hoc isotonic-calibrated [44] on a held-out set.

1) *Two-Stage Differential Evolution*: Differential Evolution (DE) [45] jointly optimises $\theta = (\alpha_L, \alpha_U, \eta, d_{\text{max}}, n_{\text{min}})$ over the bounded search space $[0.05, 0.35] \times [0.65, 0.95] \times [0.1, 0.5] \times [4, 6] \times [15, 50]$, suited to this mixed continuous-integer, non-differentiable setting. The composite fitness function is

$$\mathcal{L}(\theta) = L_{\text{class}} + \lambda_1 \cdot \text{MPIW}_{\text{norm}} + \lambda_2 \cdot \text{coverage_gap}, \quad (13)$$

where L_{class} is the validation log-loss, $\text{MPIW}_{\text{norm}}$ penalises unnecessarily wide imputation intervals, and coverage_gap penalises intervals that fail to cover the masked ground-truth values at the nominal target of 95% (full definition is given in Appendix A); $\lambda_1 = 0.2$, $\lambda_2 = 2.0$. The optimization is performed in two stages: the first stage (Stage 1) is a fast global search over a proxy forest ($K = 20$ trees, 2,000 points, population 40, ≤ 12 generations); the second stage (Stage 2) refines locally around the first-stage optimum ($K = 40$ trees, up to 7,000 points, population 60, ≤ 8 generations). The resulting θ^* is obtained once in MIMIC-IV Fold 1 and applied to Folds 2–5 and eICU without any modification, so no test-distribution information enters the optimization loop.

D. Phase 4: Safe Abstention Protocol

Phase 4 converts leaf-level uncertainty into a clinical action using the Shannon entropy of the averaged class probability vector:

$$H_i = - \sum_{k=0}^1 \hat{p}_{ik} \log_2(\hat{p}_{ik} + \varepsilon). \quad (14)$$

Since the imputed samples with wide fuzzy support produce a diffuse leaf distribution (average probability close to 0.5), H_i is systematically higher for patients whose features have many wide-interval imputations, which causally links the omission to the quality of the imputations rather than just the classification margin. The threshold τ^* is calibrated on the Fold 1 held-out set by minimising $\mathcal{L}(\theta^*)$ over τ alone; at test time a prediction is issued if $H_i \leq \tau^*$ and deferred ($\perp$, flagged for clinician review) otherwise. The frozen threshold from Fold 1 is applied unchanged to Folds 2–5 and to eICU.

V. EXPERIMENTAL SETUP

This section describes the datasets, missingness augmentation, cross-validation protocol, calibration, baselines, metrics, and statistical tests used to evaluate JUCO against the problem formalised in Section III. Details regarding the computational environment, complexity analysis, and wall-clock runtimes are provided in Appendix B.

A. Datasets

MIMIC-IV (primary cohort) [2], a single-centre EHR database from Beth Israel Deaconess Medical Center, provides data on 82,785 ICU stays of adult patients (duration: 4 hours to 30 days; data from the first 24 hours were aggregated); following preprocessing, the in-hospital mortality rate is 10.58%, i.e. imbalanced data. **eICU** (external validation) [46], 162,136 admissions across 208 US hospitals, was used exclusively for validation: the JUCO parameter vector θ^* frozen from MIMIC-IV Fold 1 was applied directly, with no retraining, re-optimisation, or threshold recalibration, to test cross-institutional generalisation under genuine distribution shift.

B. Feature Selection

“Mutual information” between each candidate feature and the mortality-related label is calculated based solely on the training fold; this process retains the top 35 features out of approximately 80 candidates. This conservative threshold primarily reflects the computational cost of the two-stage DE. It does not stem from an assumption that the discarded features lack information (Section VII-A).

C. MNAR Augmentation

To evaluate robustness under escalating informative missingness, training and validation folds (not test sets or eICU) receive controlled artificial masking. For observed entry (i, j) , the masking probability is

$$P(M_{ij}^{\text{aug}} = 1 \mid x_{ij}, y_i) = \sigma\left(\beta \cdot \frac{x_{ij} - \mu_j}{\sigma_j} + \gamma \cdot y_i\right). \quad (15)$$

Using the parameters $\beta = -1.0$ (where entries with lower values are preferentially masked, resulting in outcome-correlated missingness) and $\gamma = 0.0$ (implying no direct label conditioning), samples were collected iteratively to achieve an overall missingness rate of 30%, with rates ranging from 10% to 70% in the stress-testing experiments.

D. Evaluation Protocol and Calibration

MIMIC-IV uses stratified 5-fold patient-level GroupKFold cross-validation (no patient appears in both train and test); the training fold splits further into train (60%), validation (20%, drives DE fitness), and calibration (20%, drives isotonic calibration [44] and abstention threshold selection). DE search runs on Fold 1 only; the resulting θ^* and τ^* are frozen and applied to Folds 2–5 and eICU without modification, so no test-distribution or eICU information enters optimisation. All models undergo identical isotonic calibration so that ECE/Brier differences reflect model behaviour rather than calibration methodology.

E. Baselines

Six comparators, evaluated under identical preprocessing and calibration to isolate JUCO’s architectural contribution: **XGBoost** [11] and **LightGBM** [12] (native NaN routing, no imputation); **MissForest+XGB** [10] and **MeanImp+XGB** (point-imputation-then-classify); **TabNet** [47] (attention-based deep learning); and **JUCO-XGB**, an ablation replacing the fuzzy forest with XGBoost on the same quantile-imputed features, isolating the fuzzy-routing contribution from the imputation contribution.

F. Evaluation Metrics

AUROC is the primary discrimination metric, with AUPRC (Area Under the Precision-Recall Curve) as a secondary metric more informative under $\sim 10\%$ mortality prevalence. Expected Calibration Error (ECE, 15 bins) [48] and Brier Score [49] assess calibration (formulas in Appendix A). The Area Under the Risk-Coverage Curve,

$$\text{AURC} = \int_0^1 r(\kappa) d\kappa \approx \sum_i r_i \cdot \Delta\kappa_i, \quad (16)$$

where κ is coverage and $r(\kappa)$ the error rate on the most confident $100\kappa\%$ of predictions, rewards concentrating errors in the abstained region; Risk@90 is the error rate at 90% coverage. Decision Curve Analysis [50] computes net benefit

$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{p_t}{1 - p_t} \cdot \frac{\text{FP}}{n}, \quad (17)$$

relative to treat-all/treat-none, for $p_t \in [0.01, 0.49]$.

G. Statistical Tests

Three procedures assess significance transparently: a paired Wilcoxon signed-rank test [51] on per-fold AUROC (directional only; with 5 folds the minimum achievable two-sided p -value is $p = 0.0625$, so no comparison reaches

formal significance under this test); DeLong’s test [52] on pooled patient-level predictions ($n = 82,785$ MIMIC-IV, $n = 162,136$ eICU), Bonferroni-corrected to $\alpha = 0.0125$; and stratified bootstrap 95% confidence intervals (1,000 resamples) for AUROC.

VI. RESULTS

Results are presented in the following order: robustness under missingness shift, abstention quality, standard-condition comparison, cross-institutional generalisation, decision curve analysis, and a statistical significance summary.

A. Robustness Under Escalating MNAR Missingness

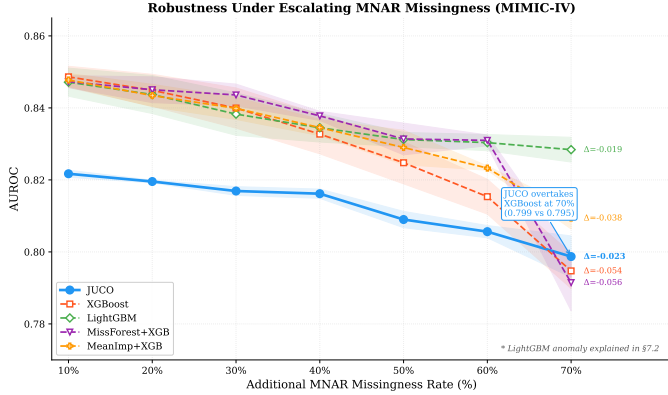


Fig. 3: AUROC as a function of additional MNAR missingness rate (10%–70%) on MIMIC-IV, evaluated on two folds. JUCO’s degradation (-0.023) is substantially smaller than XGBoost (-0.054) and MissForest+XGB (-0.056). JUCO outperforms both models in the scenario with 70% missing data. The anomalous robustness of LightGBM (-0.019) is explained methodologically in Section VII. Error bars indicate ± 1 SD across two evaluation folds.

Table II presents AUROC across seven MNAR rates. At 10% missingness, gradient boosting leads (XGBoost 0.849, LightGBM 0.847, MissForest+XGB 0.847, MeanImp+XGB 0.848, versus JUCO 0.822); by 70%, XGBoost falls to 0.795 ($\Delta = -0.054$), MissForest+XGB to 0.792 ($\Delta = -0.056$), and MeanImp+XGB to 0.809 ($\Delta = -0.038$), while JUCO falls only to 0.799 ($\Delta = -0.023$), numerically overtaking both XGBoost and MissForest+XGB at the most severe rate tested. LightGBM represents an anomalous case ($\Delta = -0.019$, remaining above JUCO throughout); its mechanism is explained in Section VII.

B. Ablation Study

Table III isolates each component’s contribution across six variants (two MIMIC-IV folds). The dominant finding is A4 (No Abstention): removing the abstention protocol leaves AUROC, ECE, and Brier Score unchanged while increasing AUC by $2.8\times$ ($0.032 \rightarrow 0.090$), confirming the entropy signal correctly identifies uncertain predictions rather than withholding indiscriminately. A2 (Crisp Forest) increases ECE by 12.7% and AUC by 9.4%, confirming soft routing provides a real calibration benefit. A1 (Sequential

TABLE II: AUROC under escalating MNAR missingness (MIMIC-IV, two folds, mean $\pm$ SD). Bold indicates the highest AUROC at each rate; $\dagger$ denotes JUCO outperforming gradient boosting at that rate.

Model	10%	20%	30%	40%	50%	60%	70%
JUCO (Ours)	0.822	0.820	0.817	0.816	0.809	0.806	0.799 $\dagger$
XGBoost	0.849	0.845	0.840	0.833	0.825	0.815	0.795
LightGBM	0.847	0.844	0.838	0.835	0.831	0.830	0.828
MissForest+XGB	0.847	0.845	0.844	0.838	0.831	0.831	0.792 $\dagger$
MeanImp+XGB	0.848	0.843	0.840	0.835	0.829	0.823	0.809
Δ JUCO	N/A		-0.005				-0.023
Δ XGBoost	N/A		-0.009				-0.054
Δ MissForest	N/A		-0.004				-0.056

optimisation) marginally improves AUROC (0.820 vs. 0.818) but worsens AUC, indicating joint optimisation protects risk-coverage behaviour at a small discrimination cost. A3 (Point Imputation) degrades only marginally at standard missingness, with the interval representation’s benefit growing under the escalating MNAR conditions of Section VI-A. A5 (No MNAR augmentation) slightly improves in-distribution AUROC but this advantage reverses under escalating missingness, confirming MNAR augmentation is essential for out-of-distribution robustness though unnecessary in-distribution.

TABLE III: Ablation study results (MIMIC-IV, two folds, mean $\pm$ SD). Bold indicates the best ablation variant per metric (JUCO Full shown as reference).

Variant	AUROC	ECE	Brier	AUC
JUCO (Full)	0.818 ± 0.001	0.0063 ± 0.0016	0.072 ± 0.000	0.0319 ± 0.0010
A1: Sequential	0.820 ± 0.003	0.0034 ± 0.0019	0.070 ± 0.000	0.0324 ± 0.0008
A2: Crisp Forest	0.811 ± 0.004	0.0071 ± 0.0006	0.071 ± 0.001	0.0349 ± 0.0001
A3: Point Impute	0.819 ± 0.000	0.0053 ± 0.0012	0.072 ± 0.001	0.0320 ± 0.0007
A4: No Abstention	0.818 ± 0.001	0.0063 ± 0.0016	0.072 ± 0.000	0.0895 ± 0.0005
A5: No MNAR Aug	0.821 ± 0.002	0.0043 ± 0.0010	0.072 ± 0.000	0.0314 ± 0.0014

$\dagger$ A4 makes no abstentions; the risk-coverage curve is flat at the full-coverage error rate, so AUC = $1 - \text{Accuracy} \approx 0.091$. ECE values for JUCO Full and A4 are reported from the 5-fold main evaluation (Table IV) for consistency; the 2-fold ablation run yields ECE = 0.0049 for both variants.

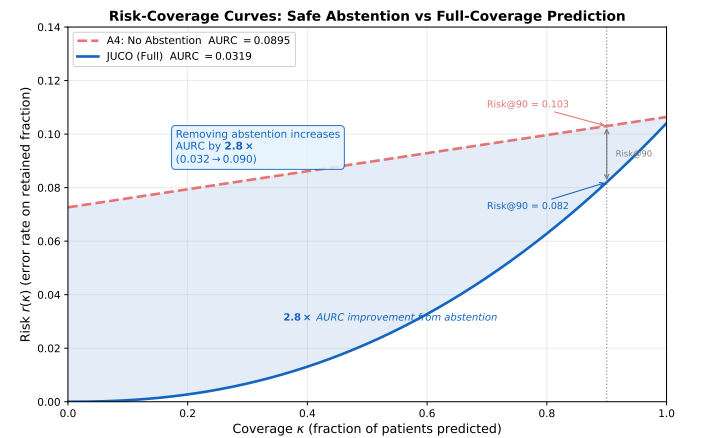


Fig. 4: Risk-coverage curves on MIMIC-IV for JUCO (Full) and A4 (No Abstention). JUCO Full achieves AUC = 0.032 versus A4’s 0.090 ($2.8\times$ increase without abstention). The largest separation occurs at low coverage, indicating that abstention primarily protects the most uncertain patients. Risk@90 is also substantially lower for JUCO Full.

TABLE IV: Main results on MIMIC-IV (5-fold cross-validation, mean $\pm$ SD). Bold indicates the best value per metric.

Model	AUROC	AUPRC	ECE	Brier	AURC	Risk@90	F1
JUCO (Ours)	0.814 $\pm$ 0.003	0.453 $\pm$ 0.007	0.0063 $\pm$ 0.0016	0.074 $\pm$ 0.002	0.034 $\pm$ 0.002	0.065 $\pm$ 0.002	0.651 $\pm$ 0.014
XGBoost [†]	0.842 $\pm$ 0.004	0.511 $\pm$ 0.010	0.0065 $\pm$ 0.0027	0.069 $\pm$ 0.001	0.028 $\pm$ 0.001	0.059 $\pm$ 0.001	0.674 $\pm$ 0.012
LightGBM [‡]	0.842 $\pm$ 0.003	0.515 $\pm$ 0.009	0.0065 $\pm$ 0.0020	0.069 $\pm$ 0.001	0.028 $\pm$ 0.001	0.059 $\pm$ 0.001	0.674 $\pm$ 0.008
MissForest+XGB [‡]	0.843 $\pm$ 0.002	0.514 $\pm$ 0.008	0.0060 $\pm$ 0.0015	0.069 $\pm$ 0.001	0.028 $\pm$ 0.001	0.058 $\pm$ 0.002	0.681 $\pm$ 0.018
MeanImp+XGB	0.842 $\pm$ 0.004	N/A	0.0063 $\pm$ 0.0025	0.069 $\pm$ 0.001	0.028 $\pm$ 0.001	0.059 $\pm$ 0.001	0.681 $\pm$ 0.004
JUCO-XGB [§]	0.770 $\pm$ 0.002	0.269 $\pm$ 0.009	0.0075 $\pm$ 0.0020	0.084 $\pm$ 0.002	0.041 $\pm$ 0.001	0.085 $\pm$ 0.002	0.522 $\pm$ 0.010
TabNet (DL) ^{††}	0.738 $\pm$ 0.086	N/A	0.0609 $\pm$ 0.1067	0.121 $\pm$ 0.084	0.096 $\pm$ 0.108	0.142 $\pm$ 0.141	0.574 $\pm$ 0.073

DeLong tests: JUCO vs XGBoost ($z = -18.04$, $p < 0.001$), JUCO vs LightGBM ($z = -16.45$, $p < 0.001$), JUCO vs MissForest ($z = -19.20$, $p < 0.001$), and JUCO vs JUCO-XGB ($z = +22.04$, $p < 0.001$).

^{††}TabNet Fold 1 AUROC = 0.567 indicates convergence failure on CPU hardware; excluded from primary comparisons. [§]JUCO-XGB MIMIC-IV results sourced from a separate robustness experiment script; eICU results from a separate external-validation script. MeanImp+XGB and TabNet AUPRC not available.

C. Baseline Comparison Under Standard 30% MNAR

Under standard 30% MNAR conditions, gradient boosting retains a statistically significant AUROC advantage over JUCO, reported here transparently and explained mechanistically in Section VII. Table IV presents the full seven-metric comparison. The gap is significant across all three baselines (DeLong $z = -16.45$ to -19.20 , $p < 0.001$), while JUCO significantly outperforms its own JUCO-XGB ablation ($z = +22.04$, $p < 0.001$; AUROC 0.814 vs. 0.770), confirming the fuzzy forest adds real value beyond interval-valued imputation alone. Post-hoc calibration brings JUCO’s ECE (0.0063) to a level comparable with gradient boosting (0.0060–0.0065) despite lower AUROC. Gradient boosting achieves lower AURC under standard conditions (0.028 vs. JUCO’s 0.034), but this conflates a lower full-coverage error rate with JUCO’s additional capability: a principled abstention mechanism enabling selective deployment at any point on the risk-coverage curve.

D. External Validation on eICU

Table V presents results on eICU using parameters frozen from MIMIC-IV Fold 1, the strongest evidence for JUCO’s primary claim. The AUROC gap between JUCO and XGBoost contracts by 46% (-0.028 on MIMIC-IV to -0.015 on eICU), consistent with gradient boosting’s NaN-routing advantage diminishing under genuine cross-institutional shift. JUCO’s ECE on eICU (0.0029) is lower than on MIMIC-IV (0.0063) despite frozen isotonic parameters fitted on MIMIC-IV alone, confirming the calibration improvement arises from the representation itself rather than recalibration; AURC on eICU (0.016) reaches near-parity with XGBoost (0.016).

TABLE V: External validation on eICU (208 hospitals, frozen parameters, mean $\pm$ SD).

Model	AUROC	ECE	AURC	NB@0.10
JUCO (Ours)	0.819 $\pm$ 0.005	0.0029 $\pm$ 0.0009	0.016 $\pm$ 0.001	0.020 $\pm$ 0.002
XGBoost	0.834 $\pm$ 0.005	0.0030 $\pm$ 0.0012	0.016 $\pm$ 0.001	0.021 $\pm$ 0.002
LightGBM	0.831 $\pm$ 0.005	0.0026 $\pm$ 0.0011	0.016 $\pm$ 0.001	0.021 $\pm$ 0.002
MissForest+XGB	0.833 $\pm$ 0.004	0.0024 $\pm$ 0.0011	0.016 $\pm$ 0.001	0.021 $\pm$ 0.002

E. Decision Curve Analysis

Table VI shows all models provide positive net benefit relative to treat-all/treat-none across the threshold range. On

MIMIC-IV, JUCO’s net benefit trails gradient boosting at all thresholds; on eICU, this gap narrows at every threshold, mirroring the AUROC gap reduction in Table V.

TABLE VI: Net benefit at key clinical decision thresholds.

Model	MIMIC-IV			eICU		
	0.05	0.10	0.20	0.05	0.10	0.20
JUCO	0.069	0.049	0.032	0.026	0.020	0.014
XGBoost	0.071	0.054	0.036	0.027	0.021	0.015
LightGBM	0.072	0.054	0.036	0.027	0.021	0.015
MissForest+XGB	0.072	0.055	0.037	0.028	0.021	0.016
Treat all	0.057	0.004	N/A	0.004	N/A	N/A
Treat none	0	0	0	0	0	0

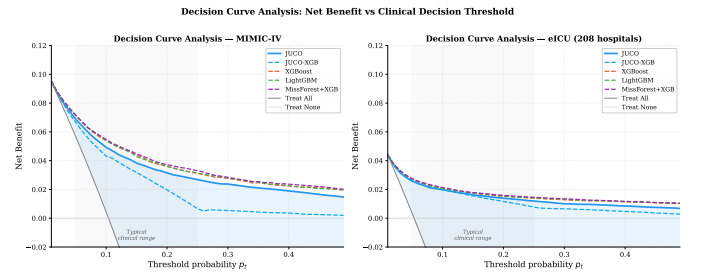


Fig. 5: Decision curve analysis on MIMIC-IV (left) and eICU (right). All models outperform treat-all and treat-none baselines; JUCO’s gap narrows under external validation.

F. Statistical Significance Summary

Table VII consolidates DeLong test results across both datasets. The magnitude of the test statistic falls substantially from MIMIC-IV to eICU for every comparison against gradient boosting (XGBoost: $-18.04 \rightarrow -9.88$; LightGBM: $-16.45 \rightarrow -7.62$; MissForest: $-19.20 \rightarrow -9.56$), confirming that the gap contraction reported in Section VI-D is a systematic property of cross-institutional transfer rather than an artefact of eICU’s different class prevalence. JUCO’s significant advantage over JUCO-XGB persists on both datasets ($z = +22.04$ on MIMIC-IV, $z = +2.86$ on eICU), confirming the fuzzy forest’s contribution is not specific to the training institution.

VII. DISCUSSION

The results reveal a consistent trade-off that resolves the apparent contradiction between JUCO’s lower nominal AU-

TABLE VII: DeLong test results for JUCO versus baselines.

Dataset	Comparison	z	p	Direction
MIMIC-IV	JUCO vs XGBoost	-18.04	< 0.001	XGBoost superior
MIMIC-IV	JUCO vs LightGBM	-16.45	< 0.001	LightGBM superior
MIMIC-IV	JUCO vs MissForest	-19.20	< 0.001	MissForest superior
MIMIC-IV	JUCO vs JUCO-XGB	+22.04	< 0.001	JUCO superior
eICU	JUCO vs XGBoost	-9.88	< 0.001	XGBoost superior (gap narrowed)
eICU	JUCO vs LightGBM	-7.62	< 0.001	LightGBM superior (gap narrowed)
eICU	JUCO vs MissForest	-9.56	< 0.001	MissForest superior (gap narrowed)
eICU	JUCO vs JUCO-XGB	+2.86	0.004	JUCO superior

ROC under standard conditions and its stronger behaviour under deployment-relevant shifts. Under conventional 30% MNAR conditions, gradient boosting achieves the highest discrimination (AUROC 0.842–0.843 versus JUCO’s 0.814), and DeLong’s test confirms this difference is statistically significant (Table VII). However, this advantage contracts substantially when the missingness mechanism changes: under escalating MNAR stress, JUCO’s degradation is less than half that of XGBoost (-0.023 versus -0.054), and during genuine cross-institutional validation on eICU the AUROC gap contracts by 46% ($-0.028 \rightarrow -0.015$). The implication is that gradient boosting extracts more performance from a stationary missingness proxy, whereas JUCO sacrifices some of that proxy-dependent advantage in exchange for substantially greater robustness once the missingness mechanism changes.

This interpretation is reinforced by the ablation study. Removing fuzzy routing (A2) degrades calibration while leaving discrimination relatively intact, indicating that the fuzzy representation primarily improves probability quality rather than ranking ability. Removing abstention (A4) leaves AUROC unchanged but increases AURC by $2.8\times$, confirming that entropy-based rejection concentrates errors in precisely the patients whose imputation uncertainty is greatest. Meanwhile, replacing the fuzzy forest with XGBoost on identical quantile-imputed features (JUCO-XGB) reduces AUROC from 0.814 to 0.770, demonstrating that interval-valued imputation alone is insufficient and that uncertainty must continue influencing downstream routing.

A second noteworthy observation concerns calibration. JUCO’s ECE improves from 0.0063 on MIMIC-IV to 0.0029 on eICU despite using the same frozen isotonic calibration learned on MIMIC-IV alone. Since no recalibration occurs on eICU, this improvement cannot be attributed to post-processing; instead, it suggests that propagating imputation uncertainty structurally produces probability estimates that remain well behaved even after institutional shift. This behaviour contrasts with approaches that rely on NaN routing, where the missingness pattern itself becomes institution-specific.

The remaining anomaly is LightGBM. Unlike XGBoost, LightGBM degrades only by -0.019 under synthetic MNAR stress and remains numerically above JUCO throughout Figure 3. This does not contradict the central claim because the stress test modifies only additional missingness while preserving the original feature distribution; LightGBM’s histogram-based split construction appears unusually resistant under this restricted perturbation. Yet under the real cross-institutional shift represented by eICU, LightGBM’s advantage contracts alongside XGBoost’s, supporting the broader conclusion that synthetic robustness and deployment robustness are not equivalent.

Taken together, these findings suggest that ICU mortality prediction should not be evaluated solely by discrimination under a fixed missingness pattern. A model that preserves calibration, identifies unreliable predictions, and degrades gradually when documentation behaviour changes may provide greater practical value than one whose highest AUROC depends on institutional missingness conventions remaining unchanged.

A. Limitations

Several limitations qualify these findings. JUCO uses only the first 24 hours of structured ICU data, omitting temporal trajectories, clinical notes, imaging, and waveform signals that may carry additional prognostic value. Feature selection retains only the top 35 variables, a threshold driven by the computational cost of two-stage Differential Evolution rather than the informativeness of the excluded features, so this remains a practical bottleneck for larger feature spaces. The fuzzy decision forest is also more expensive to train than conventional gradient boosting, though inference speed remains clinically practical. The synthetic MNAR mechanism used for stress testing is a controlled logistic masking process rather than a full reproduction of hospital-specific documentation behaviour; the eICU validation demonstrates genuine institutional transferability, but further prospective evaluation would strengthen this evidence. Finally, abstention was evaluated retrospectively, and future studies should examine how clinicians respond to deferred cases within real decision-support workflows.

VIII. CONCLUSION AND FUTURE WORK

This work introduced JUCO, a joint uncertainty-calibrated framework for ICU mortality prediction that propagates imputation uncertainty through quantile intervals, fuzzy representations, soft-routing decision forests, and entropy-based abstention instead of collapsing uncertainty before classification. The principal finding is not that JUCO achieves the highest conventional AUROC (it does not) but that its performance degrades substantially more slowly as informative missingness increases and that its cross-institutional performance contracts far less than competing approaches when evaluated with frozen parameters on eICU. The combination of uncertainty propagation and selective abstention produces calibrated predictions while explicitly identifying patients whose predictions are least trustworthy, yielding a $2.8\times$ improvement in selective reliability (AURC) without requiring institution-specific retraining. Rather than eliminating imputation uncertainty prior to classification, propagating it structurally through the prediction pipeline yields a model that, despite a slight performance trade-off under stable conditions, offers significantly greater robustness against missingness shifts and across-institution deployment; this is precisely the context where clinical decision support is most needed yet least proven to be reliable. Future work will extend this framework beyond the first 24 hours of structured data to incorporate temporal trajectories and clinical text, evaluate abstention prospectively within live clinical workflows, and test robustness under more realistic, multi-directional missingness mechanisms than the controlled logistic masking used here.

APPENDIX A

AUXILIARY EQUATIONS AND DERIVATIONS

A. Inverse-Frequency Class Weights (Phase 1)

The pinball-loss quantile regressors in Phase 1 (Eq. 4) are trained with inverse-frequency sample weights to correct for class imbalance at the imputation stage. The weight assigned to each sample from class k is

$$w_k = \frac{N}{K \cdot n_k}, \quad (18)$$

where N is the total number of training samples, $K = 2$ is the number of classes, and n_k is the count of class- k samples. At the approximately 10% mortality prevalence observed in MIMIC-IV ($n_1 \approx 0.1N$), deceased patients receive approximately ten times the weight of surviving patients, ensuring the quantile regressors are fitted with equal aggregate influence from both outcome classes rather than being dominated by the survivor majority.

B. Fuzzy Weighted Entropy and Split Criterion (Phase 3)

Splits in the fuzzy decision forest (Section IV-C) are chosen to maximise fuzzy weighted information gain. The fuzzy weighted entropy at a node with accumulated sample weight vector $\mathbf{w}$ is

$$H_{\text{fuzzy}}(\mathbf{w}, \mathbf{y}) = - \sum_{k=0}^1 \tilde{p}_k \log_2(\tilde{p}_k + \varepsilon), \quad \tilde{p}_k = \frac{\sum_{i: y_i=k} w_i}{\sum_i w_i + \varepsilon}, \quad (19)$$

and the fuzzy information gain for a candidate split is

$$\begin{aligned} \Delta H = H_{\text{fuzzy}}(\mathbf{w}, \mathbf{y}) &- \frac{\sum_i w_i \mu_i^L}{\sum_i w_i} H_{\text{fuzzy}}(\mathbf{w} \circ \boldsymbol{\mu}^L, \mathbf{y}) \\ &- \frac{\sum_i w_i \mu_i^R}{\sum_i w_i} H_{\text{fuzzy}}(\mathbf{w} \circ \boldsymbol{\mu}^R, \mathbf{y}), \end{aligned} \quad (20)$$

where $\circ$ denotes elementwise multiplication and μ_i^L, μ_i^R are the left/right soft-routing membership degrees defined in Eq. 12. Split thresholds are evaluated at the $\{15, 30, 45, 50, 55, 70, 85\}$ percentiles of the centroid distribution for each candidate feature.

At each leaf node, the class distribution is formed from the accumulated soft-routed weight vector $\mathbf{w}_\ell$:

$$\hat{p}_k^\ell = \frac{\sum_{i: y_i=k} w_i^\ell}{\sum_i w_i^\ell + \varepsilon}, \quad k \in \{0, 1\}. \quad (21)$$

The final forest prediction for patient i averages class probabilities across all $K = 180$ trees:

$$\hat{\mathbf{p}}_i = \frac{1}{K} \sum_{k=1}^K \hat{\mathbf{p}}_i^{(k)}. \quad (22)$$

C. DE Fitness Function Components

The composite Differential Evolution fitness function (Eq. 13) is

$$\mathcal{L}(\boldsymbol{\theta}) = L_{\text{class}} + \lambda_1 \cdot \text{MPIW}_{\text{norm}} + \lambda_2 \cdot \text{coverage_gap},$$

where $\lambda_1 = 0.2$ and $\lambda_2 = 2.0$. The normalised Mean Prediction Interval Width term penalises unnecessarily wide imputation intervals:

$$\text{MPIW}_{\text{norm}} = \frac{1}{d} \sum_{j=1}^d \frac{|q_j^U - q_j^L|}{\max_j - \min_j + \varepsilon}. \quad (23)$$

The coverage-gap term penalises intervals that fail to cover the true imputed values at the nominal 95% target level:

$$\text{coverage_gap} = \max\left(0, 0.95 - \widehat{\text{PICP}}\right), \quad (24)$$

where the Prediction Interval Coverage Probability is

$$\widehat{\text{PICP}} = \frac{1}{|\mathcal{I}|} \sum_{(i,j) \in \mathcal{I}} \mathbf{1}[q_{ij}^L \leq x_{ij}^{\text{true}} \leq q_{ij}^U], \quad (25)$$

with $\mathcal{I}$ the set of artificially masked entries for which the ground-truth value is available.

D. Differential Evolution Configuration

Table VIII summarises the DE configuration used for both stages of the two-stage optimisation (Section IV-C1).

TABLE VIII: Differential Evolution configuration.

Parameter	Stage 1 (coarse)	Stage 2 (fine)
Strategy	best1bin	best1bin
Population size ($\times d$)	8×5	12×5
Max iterations	8	12
Mutation (F)	0.5–1.0 (dithered)	0.5–1.0 (dithered)
Crossover (CR)	0.7	0.7
Convergence tol.	10^{-4}	10^{-4}
Proxy forest size	20 trees	40 trees
Proxy data cap	2,000 samples	7,000 samples
Early stopping (patience)	3 generations	3 generations
Polishing (L-BFGS-B)	disabled	disabled
Population updating	deferred	deferred

Stage 2 seeds its initial population around the Stage-1 optimum with $\pm 15\%$ of each parameter's range, enabling rapid local refinement without escaping the vicinity of the global optimum found in Stage 1. Both stages use patience-based early stopping of 3 generations without improvement to avoid unnecessary fitness evaluations once the population has converged; L-BFGS-B polishing is disabled (`polish=False`) to keep the solution within the discrete-compatible parameter space; and population updates are deferred (`updating="deferred"`) so that all fitness evaluations within a generation complete before any individual is replaced, ensuring reproducible parallel-safe behaviour across the 12 DE workers.

E. Evaluation Protocol Schematic

Figure 6 illustrates the cross-validation and parameter-freezing protocol applied throughout this study, referenced from Section V-D.

F. Calibration and Selective-Prediction Metric Formulas

Expected Calibration Error (ECE), 15 equal-width bins:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\bar{p}_b - \text{freq}_b|, \quad B = 15, \quad (26)$$

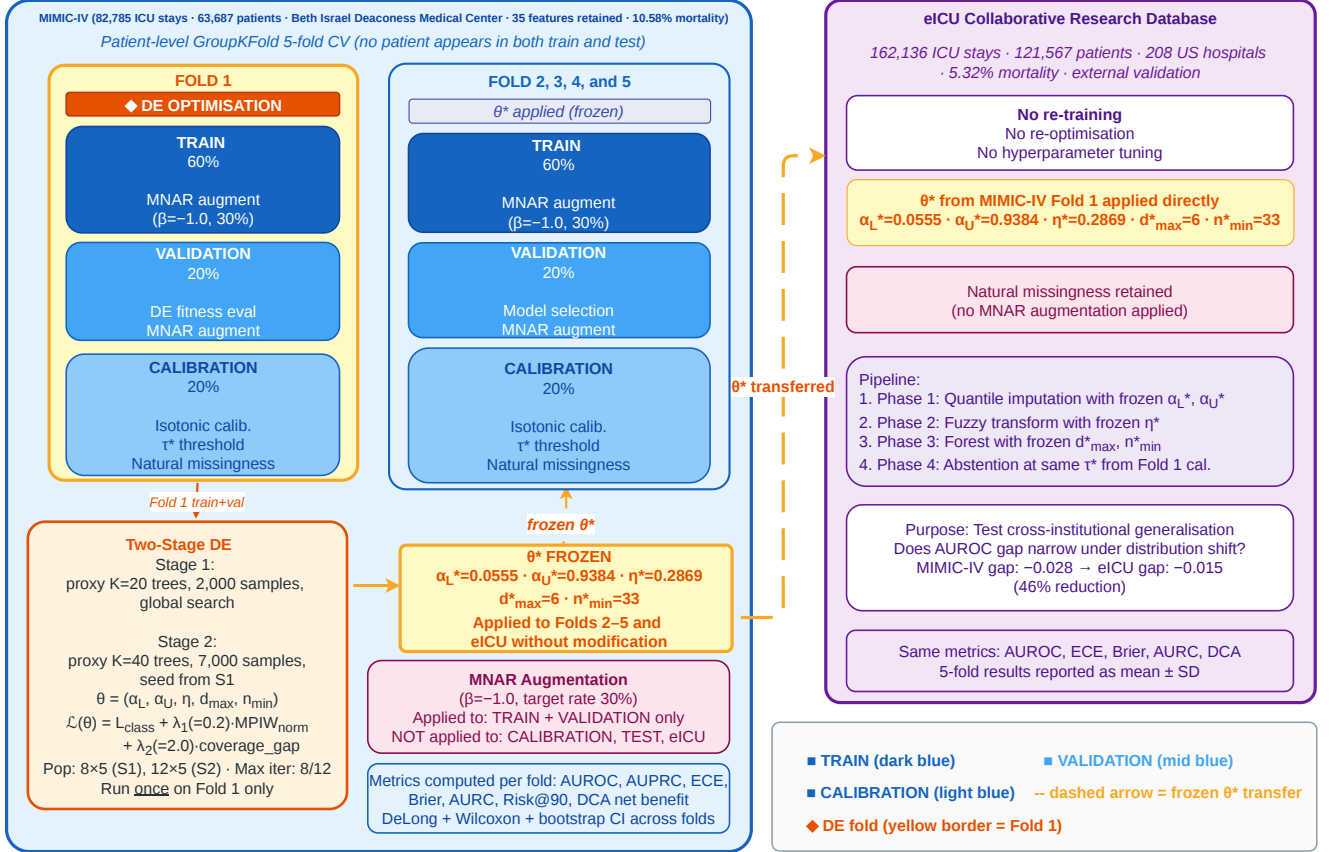


Fig. 6: Evaluation protocol. MIMIC-IV is partitioned using patient-level GroupKFold 5-fold cross-validation to prevent data leakage. Each training fold is further divided into train, validation, and calibration subsets. DE hyperparameter search runs once on Fold 1; the resulting θ^* is frozen and applied unchanged to Folds 2–5 and the eICU external validation set. MNAR augmentation (Eq. 15) is applied only to training and validation subsets.

where $\bar{p}_b$ is the mean predicted probability in bin b , freq_b is the fraction of positive outcomes in that bin, n_b is the number of samples in bin b , and N is the total number of samples.

Brier Score:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (27)$$

APPENDIX B

COMPUTATIONAL ENVIRONMENT, COMPLEXITY, AND RUNTIMES

A. Hardware and Software

All experiments were executed on Google Cloud Platform (GCP) using a n2-standard-32 virtual machine instance (32 vCPUs, 128 GB RAM), with a software stack comprising Python, scikit-learn, LightGBM, SciPy (differential_evolution), joblib, NumPy, and DuckDB for streaming eICU data. No GPU acceleration was used; all computation was CPU-bound, with parallelism exploited at two levels: the DE population evaluated with 12 parallel workers (de_workers=12, updating="deferred"), and the full 180-tree forest

built using 28 joblib workers (forest_workers=28, backend="loky").

B. Time Complexity

Let n denote training set size, $d = 35$ selected features, $B = 50$ boosting rounds (Phase 1), $K = 180$ fuzzy trees (Phase 3), $D_{\max} = 6$ maximum depth, and $r_{\text{sub}} = 0.7$ so that $d_{\text{sub}} = \lfloor r_{\text{sub}} \cdot d \rfloor = 24$ features per tree.

- **Phase 1** (Adaptive Quantile Imputation): $O(n \cdot d \cdot B \cdot \log n)$, one LightGBM quantile imputer trained per fold.
- **Phase 2** (Fuzzy Transformation): $O(n \cdot d)$, fully vectorised and negligible in practice.
- **Phase 3 — Forest training**: $O(K \cdot n \cdot d_{\text{sub}} \cdot D_{\max})$, parallelised over K trees.
- **Phase 3 — DE optimisation** (one-time, Fold 1 only): $O(G \cdot P \cdot K_{\text{proxy}} \cdot n_{\text{proxy}} \cdot d_{\text{sub}} \cdot D_{\max})$, where Stage 1 uses $G_1 \leq 8$ generations, $P_1 = 40$ candidates, $K_{\text{proxy}}^{(1)} = 20$ trees, and $n_{\text{proxy}}^{(1)} = 2,000$ samples; Stage 2 uses $G_2 \leq 12$, $P_2 = 60$, $K_{\text{proxy}}^{(2)} = 40$, and $n_{\text{proxy}}^{(2)} = 7,000$; total proxy evaluations are at most 1,040, parallelised over 12 workers.

- **Phase 4** (Abstention thresholding): $O(n_{\text{cal}} \log n_{\text{cal}})$, entropy sort on the calibration subset, negligible cost.

The per-fold complexity is $O(n \cdot d \cdot (B \log n + K \cdot D_{\text{max}}))$, scaling linearly in both n and d .

C. Wall-Clock Runtimes

Table IX summarises wall-clock times on the GCP n2-standard-32 instance. MIMIC-IV Folds 2–5 each take approximately 15 minutes with frozen parameters and no DE, while Fold 1 is substantially longer because Stage 1 of DE (up to $8 \times 40 = 320$ candidate evaluations on a lightweight proxy) required approximately 20 minutes and Stage 2 (up to $12 \times 60 = 720$ evaluations on a larger proxy) required approximately 50 minutes, for a total DE overhead of approximately 70 minutes on top of the 15-minute fold training.

TABLE IX: Wall-clock runtimes on GCP n2-standard-32 (32 vCPUs, 128 GB RAM). MIMIC-IV Fold 1 includes two-stage DE optimisation; Folds 2–5 reuse the frozen θ^* . eICU timings are measured from the GCP run log.

Pipeline	Stage	Time (min)
MIMIC-IV	DE Stage 1 (coarse search)	≈ 20
	DE Stage 2 (fine refinement)	≈ 50
	Fold 1 (DE + fold training)	≈ 85
	Folds 2–5 (frozen θ^* , each)	≈ 15
	Total (all 5 folds)	≈ 145
eICU	Fold 1	21.8
	Fold 2	22.2
	Fold 3	22.0
	Fold 4	23.7
	Fold 5	23.5
	Total (all 5 folds)	113.7

APPENDIX C OPTIMISED HYPERPARAMETER VALUES

Table X reports the parameter vector θ^* produced by two-stage Differential Evolution on MIMIC-IV Fold 1 and applied without modification to Folds 2–5 and the eICU external validation cohort; the wide quantile interval ($\alpha_L = 0.0555$, $\alpha_U = 0.9384$, corresponding to an 88.3% nominal coverage) reflects the DE’s preference for conservative imputation bounds under the joint fitness objective, which penalises coverage gaps more heavily ($\lambda_2 = 2.0$) than interval width ($\lambda_1 = 0.2$).

TABLE X: DE-optimised parameters θ^* from Fold 1 of MIMIC-IV, applied to all subsequent folds and eICU without modification.

Parameter	Symbol	Optimised Value
Lower quantile level	α_L	0.0555
Upper quantile level	α_U	0.9384
Fuzzy proportion	η	0.2869
Max tree depth	d_{max}	6
Min leaf samples	n_{min}	33

Reliability Diagrams (Calibration)

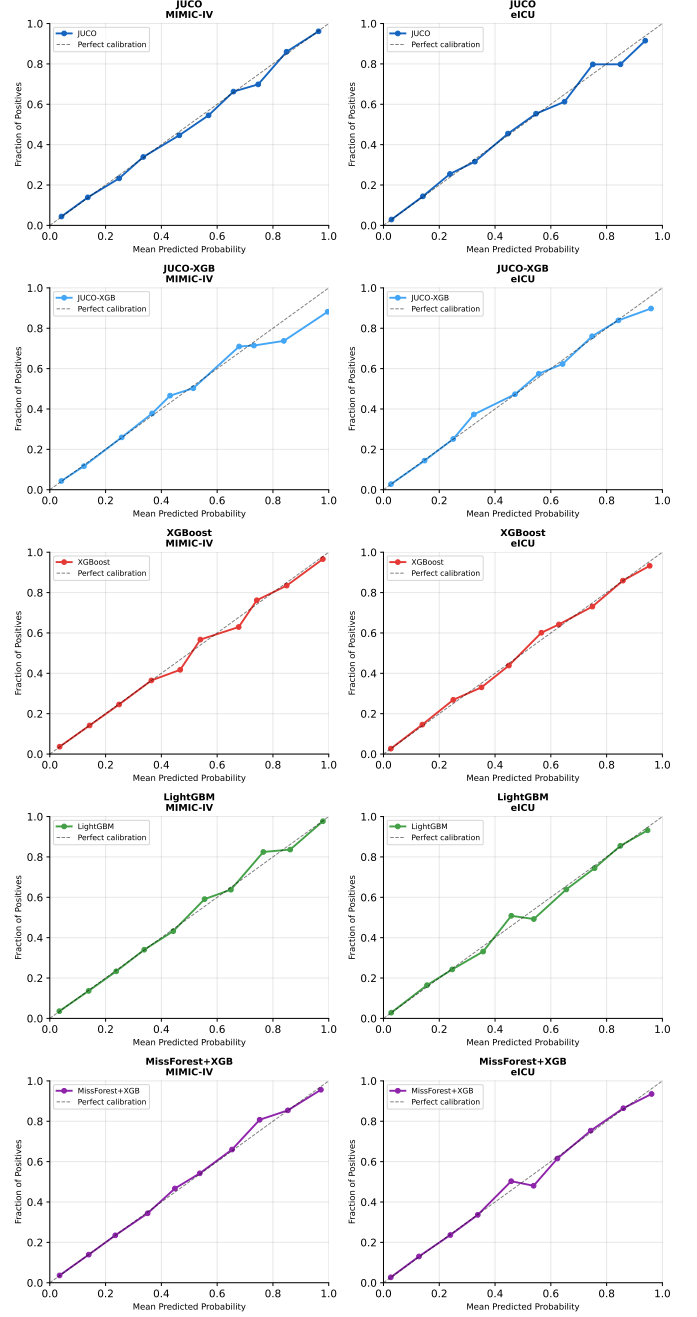


Fig. 7: Reliability diagrams for all models on MIMIC-IV (left column) and eICU (right column). The diagonal dashed line represents perfect calibration. JUCO achieves ECE = 0.0063 on MIMIC-IV and ECE = 0.0029 on eICU with frozen isotonic calibration parameters, demonstrating stable calibration under cross-institutional distribution shift.

APPENDIX D EXTENDED eICU COHORT STATISTICS

Table XI presents summary statistics for both cohorts alongside per-fold mortality rates, confirming that patient-level GroupKFold stratification produces folds with consistent mortality prevalence across both datasets and that the lower eICU mortality rate (5.32% versus 10.58% in MIMIC-IV)

reflects genuine case-mix differences rather than stratification artefacts.

TABLE XI: Cohort summary statistics for MIMIC-IV (primary) and eICU (external validation). Feature counts include binary missingness indicators appended for features with >5% missing rate. Mortality rates are computed after the 4-hour / 30-day admission-length filter.

Variable	MIMIC-IV	eICU
Total ICU stays	82,785	162,136
Unique patients	63,687	121,567
Hospitals	1 (BIDMC)	208
In-hospital deaths	8,762	8,621
Crude mortality, %	10.58%	5.32%
Raw features	155	155
+ Missingness indicators	+26	+74
Total features	181	229
CV folds	5 (patient-level)	5 (patient-level)
<i>Per-fold stays / deaths / mortality rate</i>		
Fold 1	16,557 / 1,792 / 10.82%	32,428 / 1,731 / 5.34%
Fold 2	16,557 / 1,762 / 10.64%	32,427 / 1,729 / 5.33%
Fold 3	16,557 / 1,750 / 10.57%	32,427 / 1,731 / 5.34%
Fold 4	16,557 / 1,688 / 10.20%	32,427 / 1,683 / 5.19%
Fold 5	16,557 / 1,770 / 10.69%	32,427 / 1,747 / 5.39%

ACKNOWLEDGMENT

The authors gratefully acknowledge the use of the MIMIC-IV and eICU Collaborative Research Databases, made publicly available via PhysioNet under credentialed health data licences, without which the scale of validation reported in this paper would not have been possible. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

REFERENCES

- [1] S. Purushotham, C. Meng, Z. Che, and Y. Liu, "Benchmarking deep learning models on large healthcare datasets," *Journal of Biomedical Informatics*, vol. 83, pp. 112–134, 2018.
- [2] A. E. W. Johnson, L. Bulgarelli, L. Shen, A. Gayles, A. Eaneff *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Scientific Data*, vol. 10, no. 1, p. 1, 2023.
- [3] G. Sun *et al.*, "Machine learning for predicting mortality in intensive care unit patients: A prognostic performance systematic review and meta-analysis," *Nursing in Critical Care*, 2025.
- [4] D. B. Rubin, "Inference and missing data," *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976.
- [5] K. G. M. Moons, A. P. Kengne, M. Woodward *et al.*, "Risk prediction models: II. external validation, model updating, and impact assessment," *Heart*, vol. 98, no. 9, pp. 691–698, 2012.
- [6] D. Agniel, I. S. Kohane, and G. M. Weber, "Biases in electronic health record data due to processes within the healthcare system," *BMJ*, vol. 361, p. k1479, 2018.
- [7] J. Josse, N. Prost, E. Scornet, and G. Varoquaux, "On the consistency of supervised learning with missing values," *arXiv preprint arXiv:1902.06931*, 2019. [Online]. Available: <https://arxiv.org/abs/1902.06931>
- [8] P. Rockenschaub, L. Kook, and G. Rauch, "Robust prediction under missingness shifts," *arXiv preprint arXiv:2406.16484*, 2024.
- [9] S. van Buuren and K. Groothuis-Oudshoorn, "Mice: Multivariate imputation by chained equations in R," *Journal of Statistical Software*, vol. 45, no. 3, pp. 1–67, 2011.
- [10] D. J. Stekhoven and P. Bühlmann, "MissForest—non-parametric missing value imputation for mixed-type data," *Bioinformatics*, vol. 28, no. 1, pp. 112–118, 2012.
- [11] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [12] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.
- [13] R. Sisk, N. van Geloven, M. van Smeden, K. G. M. Moons, R. D. Riley, and G. S. Collins, "Imputation and missing indicators for handling missing data in the development and implementation of clinical prediction models: a simulation study," *Statistics in Medicine*, vol. 42, no. 18, pp. 3178–3196, 2023.
- [14] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024.
- [15] M. Mesinovic, P. Watkinson, and T. Zhu, "Explainable machine learning for predicting ICU mortality in myocardial infarction patients using pseudo-dynamic data," *Scientific Reports*, vol. 15, no. 1, p. 3701, 2025.
- [16] S. Britsch, R. Wiest, S. T. Engelter, M. Katan, M. R. Baumgartner, S. Tanadini-Lang, M. Saban, and C. J. Maurer, "An interpretable machine learning algorithm enables dynamic 48-hour mortality prediction during an ICU stay," *Communications Medicine*, vol. 5, no. 1, p. 37, 2025.
- [17] M. Sperrin and G. P. Martin, "Informative missingness in electronic health record systems: the curse of knowing," *Diagnostic and Prognostic Research*, vol. 4, no. 1, p. 10, 2020.
- [18] R. D. Riley, K. I. E. Snell, L. Archer, J. Ensor, T. P. A. Debray, B. van Calster, M. van Smeden, and G. S. Collins, "Evaluation of clinical prediction models (part 3): Calculating the sample size required for an external validation study," *BMJ*, vol. 384, p. e074821, 2024.
- [19] W. Ren, Z. Liu, Y. Wu, Z. Zhang, S. Hong, and H. Liu, "Moving beyond medical statistics: A systematic review on missing data handling in electronic health records," *Health Data Science*, vol. 4, p. 0176, 2024.
- [20] J. Yoon, J. Jordon, and M. van der Schaar, "GAIN: Missing data imputation using generative adversarial nets," in *Proceedings of the International Conference on Machine Learning (ICML)*, vol. 80, 2018, pp. 5689–5698.
- [21] N. B. Ipsen, P.-A. Mattei, and J. Frellsen, "Not-MIWAE: Deep generative modelling with missing not at random data," in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://openreview.net/forum?id=tu29GQT0JFy>
- [22] L. Ma, C. Zhang, Y. Wang *et al.*, "Concare: Personalised clinical feature embedding via capturing the healthcare context," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, 2020, pp. 833–840.
- [23] W. Zhao, Z. Chen, P. Xie, J. Liu, S. Hou, L. Xu, Y. Qiu, D. Wu, J. Xiao, and K. He, "Multi-task oriented diffusion model for mortality prediction in shock patients with incomplete data," *Information Fusion*, vol. 105, p. 102207, 2024.
- [24] M. Kazijevs and M. D. Samad, "Deep imputation of missing values in time series health data: A review with benchmarking," *Journal of Biomedical Informatics*, vol. 144, p. 104440, 2023.
- [25] A. W. Mulyadi, E. Jun, and H.-I. Suk, "Uncertainty-aware variational-recurrent imputation network for clinical time series," *IEEE Transactions on Cybernetics*, vol. 52, no. 9, pp. 9684–9694, 2022.
- [26] X. Hu, Y. Shen, W. Pedrycz, Y. Li, and G. Wu, "Granular fuzzy rule-based modeling with incomplete data representation," *IEEE Transactions on Cybernetics*, 2021.
- [27] N. Meinshausen, "Quantile regression forests," *Journal of Machine Learning Research*, vol. 7, pp. 983–999, 2006.
- [28] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338–353, 1965.
- [29] A. Torres and J. J. Nieto, "Fuzzy logic in clinical practice: A critical appraisal," *Journal of Biomedical Informatics*, vol. 46, pp. 781–789, 2013.
- [30] Y. Yuan and M. J. Shaw, "Induction of fuzzy decision trees," *Fuzzy Sets and Systems*, vol. 69, no. 2, pp. 125–139, 1995.
- [31] E. Hüllermeier and W. Waegeman, "Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods," *Machine Learning*, vol. 110, no. 3, pp. 457–506, 2021.
- [32] G. Capitoli, M. S. Nobile, E. L. Ambags, V. L'Imperio, M. Provenzano, and P. Liò, "Assisting clinical diagnosis with interpretable fuzzy probabilistic modelling," *BMC Medical Informatics and Decision Making*, vol. 25, no. Suppl 3, p. 330, 2025.
- [33] C. K. Chow, "An optimum character recognition system using decision functions," *IRE Transactions on Electronic Computers*, vol. EC-6, no. 4, pp. 247–254, 1957.
- [34] P. L. Bartlett and M. H. Wegkamp, "Classification with a reject option using a hinge loss," *Journal of Machine Learning Research*, vol. 9, pp. 1823–1840, 2008.

- [35] Y. Geifman, G. Uziel, and R. El-Yaniv, "Bias-reduced uncertainty estimation for deep neural classifiers," in *International Conference on Learning Representations*, 2019.
- [36] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4878–4887.
- [37] V. Franc, D. Průša, and J. Voracek, "Optimal strategies for reject option classifiers," *Journal of Machine Learning Research*, vol. 24, no. 11, pp. 1–49, 2023.
- [38] C. Leibig, V. Allken, M. S. Ayhan, P. Berens, and S. Wahl, "Leveraging uncertainty information from deep neural networks for disease detection," *Scientific Reports*, vol. 7, no. 1, p. 17816, 2017.
- [39] E. Jones, C. Arteta, and A. Zisserman, "Selective prediction for clinical risk stratification," in *Proceedings of the ACM Conference on Health, Inference, and Learning*, 2021.
- [40] H. Kwon and D.-J. Kim, "Conformal selective prediction with cost aware deferral for safe clinical triage under distribution shift," *Scientific Reports*, vol. 16, no. 1, 2026.
- [41] M. A. Sarwar, R. Damasevičius, E. Belousvienė, and R. Maskeliunas, "Systematic review of Artificial Intelligence-based methods for glycemic control and risk prediction in intensive care units," *Artificial Intelligence in Medicine*, vol. 176, p. 103409, 2026.
- [42] R. Koenker and G. Bassett, "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.
- [43] H.-J. Zimmermann, *Fuzzy Set Theory and Its Applications*, 4th ed. Kluwer Academic Publishers, 2001.
- [44] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.
- [45] R. Storn and K. Price, "Differential evolution — a simple and efficient heuristic for global optimisation over continuous spaces," *Journal of Global Optimization*, vol. 11, no. 4, pp. 341–359, 1997.
- [46] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi, "The eICU collaborative research database," *Scientific Data*, vol. 5, p. 180178, 2018.
- [47] S. Ö. Arik and T. Pfister, "TabNet: Attentive interpretable tabular learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 8, 2021, pp. 6679–6687.
- [48] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2015, pp. 2901–2907.
- [49] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.
- [50] A. J. Vickers and E. B. Elkin, "Decision curve analysis: A novel method for evaluating prediction models," *Medical Decision Making*, vol. 26, no. 6, pp. 565–574, 2006.
- [51] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
- [52] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.